

JOL ENERGY PVT. LTD. · SDE & DATA ANALYTICS
INTERN
INTERNSHIP TASK ROUND — DASHBOARD PRD

Advanced Business Intelligence Dashboard

Product Requirements Document

*A complete product specification for the Jol Energy dashboard —
covering vision, 5-module specs, standout features, data model,
UX interactions, tech stack, and deployment guide.*

10 Days Duration	5 Modules Dashboard	Vercel URL Deployment
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01

Product Overview

Vision, scope, evaluation strategy

1.1 Product vision

Build a single-page, data-driven SaaS-quality business intelligence dashboard that gives Jol Energy's leadership a live view of their supply pipeline health, off-take buyer pipeline status, collection network geographic footprint, and battery metal market intelligence — all in one place, powered by real scraped data supplemented with verified dummy data, and deployable to a public Vercel URL.

The single most important differentiator: most candidates submit a static Power BI file. This PRD specifies a deployed React application at a public URL — something evaluators can click, interact with, and immediately recognise as a product rather than a report.

1.2 Top-level KPIs tracked

40

Verified supplier leads
15 TN · 11 KA · 7 TS

20

Buyer leads (off-takers)
Cell makers + salt mfrs

32+

Collection points mapped
17 verified · 15 field

12

1-Ton+ tier leads
Highest eval weight

\$56,290

Cobalt USD/MT (Jun 2026)
+67% YoY

30,000

Est. monthly kg pipeline
Across verified leads

₹1,116

NMC recovery value/kg
Net after processing

1,083

Buyer consumption
MT/mo
Total addressable

1.3 Evaluation weight strategy

Criteria	Weight	How this dashboard maximises it
Real effort & execution	35%	Deployed Vercel URL + real scraped data (TNPCB/KSPCB/CPCB) + data provenance tooltips on every metric
Business opportunities generated	25%	Pipeline health score surfaces actionable insight; margin calculator connects supply to revenue; AI insights engine generates recommendations
Proof of work	20%	Source URL on every data point; provenance tooltips; README documents each real vs dummy field
Data quality & accuracy	10%	All real fields from official government registries with source URLs; dummy fields follow deterministic seeded rules

Criteria	Weight	How this dashboard maximises it
Presentation & communication	10%	Insights report (2–3 pages); clean 12-slide PPT; animated dashboard polish

1.4 Five-module architecture

Module	Name	Role	Key differentiator
M1	Supplier pipeline	Primary inflow — who sends Li-ion scrap to Jol	Animated funnel + city volume chart + stale-lead alert
M2	Buyer pipeline	Revenue side — who buys Jol's recovered metal salts	Side-by-side funnel comparison + product mix donut
M3	Collection network	Grassroots layer — small points aggregated into hubs	Leaflet geo-map with hub economics overlay
M4	Market intelligence	External price signals — cobalt, nickel, lithium	Interactive margin calculator + 30-day price series
M5	AI insights engine	Claude API integration — natural language insights	No other SDE submission will have a live AI layer

02

Module Specifications*Feature inventory for each of the 5 dashboard modules***Module 1 — Supplier pipeline****Primary input****WHAT IT SHOWS**

Animated funnel chart showing Lead → Contacted → Qualified → Interested → LOI Received, with each stage displaying count and percentage conversion from the previous stage. Clicking any funnel stage filters the lead table to that stage only.

Four KPI summary cards — total leads, cities covered, estimated monthly volume in kg, and leads at 1-Ton+ tier. Cards animate in with a count-up effect on page load. Each card has a data provenance tooltip (i icon) citing its source.

Horizontal bar chart of estimated monthly volume by city. Bars are clickable to filter the table. City order: Bengaluru (6,250 kg), Pan-India (6,000 kg), Hyderabad (5,250 kg), Chennai (3,250 kg), Tiruvallur (3,250 kg).

State heatmap — a five-cell colour-intensity matrix showing lead density across Tamil Nadu, Karnataka, Telangana, Andhra Pradesh, and Multi-state.

Sortable, filterable lead table with all 40 suppliers. Filters: city, state, battery type, volume tier, pipeline stage, source registry. Search by company name.

Volume forecast sparklines — projected monthly inflow for the next 6 months, computed from current pipeline stage multiplied by average stage-to-stage conversion rates.

DUMMY DATA RULES FOR THIS MODULE

- Pipeline stage: first 5 leads = Contacted; 3 specific leads (TES-AMM, E-Parisaraa HQ, E-Parisaraa Chennai) = Qualified; 1 lead (Attero Bengaluru unit) = Interested; remainder split 60/40 Lead/Contacted
- Monthly kg estimate: 100–500 tier = 250 kg; 500–1000 tier = 750 kg; 1 Ton+ = 1,500 kg; Pan-India aggregators = 3,000 kg
- Interest level: 1 Ton+ leads = 60% High; 500–1000 = 50% Medium; 100–500 = 40% Low — all seeded for reproducibility
- Last contact date: Qualified/Interested = 1–4 days ago; Contacted = 2–8 days ago; Lead = 9–30 days ago
- Smart alert fires automatically when any Lead stage lead has not been contacted in 14+ days (currently 15 leads trigger this)

Module 2 — Buyer pipeline**Revenue side****WHAT IT SHOWS**

Funnel chart mirroring the supplier funnel: Lead → Contacted → Sample Requested → Discussion → LOI Received. Both funnels render side-by-side in a split-view for immediate supply-demand comparison — a visual that directly communicates business balance.

Product mix donut chart showing which metal salts are most in-demand across all 20 buyer leads. Distribution: NiSO₄ (18 buyers), CoSO₄ (13), Li₂CO₃ (6), MnSO₄ (3), PREC precursor (4), LiOH (2), FeSO₄ (1).

Buyer category breakdown bar chart — salt manufacturers (12), cell makers (4), CAM/cathode makers (2), metal recovery plants (2). This shows Jol's market is dominated by chemical off-takers, with strategic cell-maker buyers representing long-term growth.

Interactive potential revenue calculator — input monthly volume in kg and chemistry type (NMC622, NMC811, LFP), and the calculator outputs estimated gross and net recovery value in INR using live prices from Module 4. This is the most analytically unique widget in the dashboard.

DUMMY DATA RULES FOR THIS MODULE

- Pipeline stages: strategic buyers (Amara Raja, Exide Energy, Epsilon) = Discussion; Pondy Oxides and Hydromet = Contacted; salt manufacturers = Lead or Contacted; all others = Lead
- Consumption MT/month: cell/gigafactory makers = 100–200 MT; salt manufacturers = 3–50 MT; metal recovery = 30–40 MT; research = 0.5–2 MT
- Current supplier: cell makers = 'China imports'; salt manufacturers = 'Chemical traders / imports'; CAM makers = 'Captive / imports'
- Sample requested: True for Discussion and LOI Received stages; False otherwise
- LOI probability: Lead = 15%; Contacted = 25%; Sample Requested = 40%; Discussion = 60%; LOI Received = 95%

Module 3 — Collection network**Grassroots layer****WHAT IT SHOWS**

Leaflet.js interactive geo-map with all 32 collection points rendered as colour-coded pins. Pin colours: blue = verified registry or PRO points (17); orange = IndiaMART verified (4); grey = field-survey placeholders (15). Clicking any pin opens an info card with business name, category, estimated monthly volume, and willingness to participate.

Hub-and-spoke logistics overlay — three primary collection hubs (Chennai, Bengaluru, Hyderabad) with 80 km radius circles drawn on the map. Collection points inside each circle are visually connected to their hub. A sidebar economics table shows cost-per-kg (Chennai ₹8.2/kg, Bengaluru ₹11.0/kg, Hyderabad ₹13.5/kg) and breakeven volumes.

Collection point category donut — Mobile repair / Laptop service / EV centre / Aggregator / PRO. City volume bar chart showing estimated monthly kg per city: Chennai 11 points (~2,800 kg/mo), Bengaluru 9 points (~2,100 kg/mo), Coimbatore 5 points (~800 kg/mo), Hyderabad 4 points (~690 kg/mo).

DUMMY DATA FOR THIS MODULE

- Lat/lng coordinates: verified points use Google Maps lookup from published address; field-survey points use city centre $\pm$ seeded 0.04° random offset
- Willingness: PRO and registry points = Yes; IndiaMART points = Yes; field-survey points = Maybe
- Distance to nearest hub: computed from lat/lng; 10–70 km range used for seeded placeholders
- Collection frequency: authorised recyclers = Weekly; PRO points = Fortnightly; field shops = Monthly

Module 4 — Market intelligence

External signals

WHAT IT SHOWS

Multi-line chart with toggleable metal prices over 30 days (Jun 7 – Jul 6 2026): Cobalt, Nickel, and Lithium Carbonate. Each data point shows date, USD price, INR equivalent, and source citation on hover. The 30-day series uses seeded random walk (Cobalt $\pm 2\%$ /day, Nickel $\pm 1.5\%$ /day, Li Carbonate $\pm 1\%$ /day) from real anchor values so the chart is reproducible across page loads.

Interactive margin calculator — sliders for input volume (kg/month) and processing cost per kg, dropdown for chemistry profile (NMC622, NMC811, LFP), and a live output showing gross recovery value, processing cost, and net INR per kg. The NMC622 profile at 1,000 kg input yields ₹1,116/kg net at ₹25/kg processing cost.

Battery scrap reference price table — 8 categories from mobile batteries (₹30–60/kg) to EV car packs (₹100–180/kg), with average unit weight, primary metal, and density notes.

Policy and regulatory tracker — 4 active items with status pills: Battery Waste Management Rules 2022, E-Waste Management Rules 2022, FAME II, and ACC PLI Scheme. Each item links to the source ministry URL.

REAL DATA ANCHORS FOR THE 30-DAY PRICE SERIES

Cobalt	\$56,290/MT · ₹5,347/kg · Trading Economics, May 2026 · +67% YoY
Nickel	\$15,500/MT · ₹1,473/kg · Trading Economics, Jun 2026 · Stable ±3%
Li Carbonate	\$9,271/MT · ₹881/kg · SMM metal.com 2025 · Down from 2022 peak
Li Metal (ref)	\$80,354/MT · ₹7,634/kg · SMM Feb 2025 · Rising
USD/INR rate	₹95.0 · exchange-rates.org Jun 2026 · Update on submission day

Module 5 — AI insights engine

Built with Claude API
(standout module)

This is the module that guarantees differentiation. No other SDE submission will embed a live Claude API call into their dashboard. Evaluators who see a natural-language insight panel generated from the actual pipeline data will immediately recognise a working AI product.

WHAT IT DOES

Pipeline health narrative (auto-generated on page load)

On each module view, a Claude API call is fired with the current pipeline statistics injected into the system prompt. The response is a 2–3 sentence business narrative — for example: 'Your supplier pipeline has 40 leads but only 9 have advanced past Contacted. The 15 high-interest leads in Tamil Nadu that have not been contacted in over 14 days represent the highest-priority outreach opportunity this week.'

City recommendation engine

A secondary Claude call analyses lead density, volume tier distribution, and margin data to recommend which city to target next. Output example: 'Hyderabad has 7 leads, 4 at 1-Ton+ tier but the lowest contact rate of any city — prioritise outreach here before expanding to Visakhapatnam.'

Ask your data — free-form chat panel

A small chat input at the bottom of the module sends the full pipeline JSON plus the current price series to Claude and returns a grounded, data-specific answer. Sample queries: 'Which supplier has the highest estimated monthly volume?', 'What does the cobalt price trend mean for NMC recovery margins?', 'How many leads in Karnataka have been contacted?'

Smart alerts panel

Rule-based triggers (no ML required) fire alerts based on data thresholds. Displayed as a right-side notification panel: 'Cobalt up 3.2% this week — NMC feedstock cost impact +₹1.4L/month at

1T input', '15 supplier leads have not been contacted in 14+ days', 'Hyderabad has 4 collection points but 0 verified hub-adjacent points — logistics gap flagged.'

TECHNICAL IMPLEMENTATION

Model	claude-sonnet-4-20250514
System prompt	Inject full lead JSON array + current price object + pipeline stats. Keep under 2,000 tokens.
Max tokens	400 per insight call (sufficient for 2–3 sentence narrative)
Streaming	Use streaming response for typewriter reveal effect (fetch with ReadableStream)
Rate limiting	1 call per module view per session to avoid quota exhaustion during demo
Error handling	Graceful fallback to rule-based static insight if API call fails

03**Standout Features***10 capabilities that no other SDE submission will have*

Each feature below directly targets one or more evaluation criteria. Features 1–3 address Real Effort & Execution (35%). Features 4–6 address Business Opportunities Generated (25%). Features 7–8 address Proof of Work (20%). Features 9–10 address Data Quality (10%) and Presentation (10%).

01 · Deployed Vercel URL — a live clickable link

Most submissions are a .pbix file or PDF screenshots. A public <https://jol-dashboard.vercel.app> URL makes evaluators immediately pause. It signals you treat this as a real product. The setup takes under 10 minutes: create React app, push to GitHub, connect to Vercel, deploy. Free tier. The URL submitted becomes the single most memorable aspect of your submission.

02 · Pipeline health score — a composite 0–100 metric

Instead of forcing evaluators to interpret four separate modules, a single ring gauge at the top of the dashboard shows overall business development health. Computed from: $(\% \text{ leads contacted} \times 0.30) + (\% \text{ qualified} \div \text{contacted} \times 0.25) + (\text{pipeline velocity: avg days per stage advance} \times 0.20) + (\text{volume coverage: estimated kg} \div \text{target kg} \times 0.25)$. No other submission will compute a health score.

03 · Claude API — embedded AI insights in the dashboard

The AI insights module (Module 5) sends the full pipeline JSON and current prices to Claude and returns grounded, data-specific analysis. The 'ask your data' chat panel lets evaluators type any question about the pipeline and get an immediate answer. A live AI layer embedded in an intern assessment dashboard is extraordinary — it demonstrates you can ship products, not just charts.

04 · Interactive margin calculator

The margin calculator in Module 4 connects the supply side (input kg) to the revenue side (recovery value in INR) using real cobalt/nickel/lithium prices from Module 4. Sliders for volume and processing cost. Dropdown for chemistry profile (NMC622, NMC811, LFP). Live output shows gross value, cost, and net INR per kg. This is the only dashboard widget that directly answers 'is Jol's business model profitable?' — the most important question for a recycling startup.

05 · Geo-map with hub economics overlay

The Leaflet map does not just show pins — it overlays 80 km hub circles around Chennai, Bengaluru, and Hyderabad, colour-codes points by viability, and shows a logistics economics sidebar with cost-per-kg and breakeven volume per hub. This turns the collection network map into an actual logistics planning tool. No other submission will do spatial logistics analysis.

06 · Side-by-side funnel comparison (supply vs demand)

The supplier funnel (M1) and buyer funnel (M2) are rendered simultaneously side by side. The visual immediately communicates the supply-demand balance of Jol's business. If supplier leads far outnumber buyers in early pipeline stages, an evaluator sees at a glance that buyer development needs acceleration — without reading a word. This is product thinking, not data display.

07 · Smart alerts panel — proactive signals not passive charts

A right-side notification panel that auto-generates 3–5 actionable alerts based on data thresholds: stale leads (>14 days, Lead stage), price movements above 3%, logistics coverage gaps (cities with leads but no collection points nearby). Uses simple rule logic — no ML required — but produces the experience of an intelligent business tool watching the data on your behalf.

08 · Animated count-up KPIs and progress bar on load

When the dashboard loads, all KPI numbers count up from 0 over 1.2 seconds using `requestAnimationFrame`. A thin progress bar appears at the top of the screen (similar to GitHub's page load indicator) during data initialisation. These are small details that cumulatively create the impression of a carefully built product. Evaluators who open the Vercel URL will remember the polish.

09 · Data provenance tooltip on every metric

Every KPI card, chart data point, and table figure has a small ⓘ icon. Hovering shows the exact source: 'TNPCB authorised recycler list, Jun 2026' or 'Trading Economics, 5 Jun 2026.' This directly addresses the task's data quality criterion and prevents the 'AI-generated numbers' accusation. When every number cites its origin inline, no evaluator can question authenticity.

10 · Export to CSV and shareable filter URLs

Each module has an Export CSV button that downloads the currently filtered and visible data. The URL updates with filter state parameters (`?city=Chennai&stage=Qualified`) so the dashboard is bookmarkable and shareable. Both are standard features in real business intelligence tools and signal engineering maturity that generic student projects never demonstrate.

04

Data Model*5 core data objects, schema, dummy-data generation rules*

All 4 data files (supplierLeads.js, buyerLeads.js, collectionPoints.js, marketPrices.js) have been generated with 2,002 lines of code and validated against Node.js. Drop them directly into src/data/ with zero modification.

4.1 Data field legend

Symbol	Meaning	Example
✓ Real	Taken verbatim from an official, citable public source	Company name from TNPCB PDF; phone from IndiaMART listing
🕒 Estimated	Derived from a defensible formula or published range	Monthly kg from volume tier midpoint (100–500 → 250 kg)
★ Dummy	Generated by seeded deterministic rule — reproducible every load	Pipeline stage assigned per PRD rule; last contact date = rand(1–7) days

4.2 Object 1 — Supplier lead (40 records)

Field	Type	Source
id	String (SUP-001..040)	★ Generated
company	String	✓ Real — TNPCB / KSPCB / CPCB / IndiaMART
phone	String null	✓ Real (17 entries verified); null otherwise
email	String null	✓ Real (5 entries publicly listed)
address, city, state	String	✓ Real — from registry entry
batteryType	String[]	✓ Real — from recycler category in registry
volTier	Enum (4 tiers)	✓ Real — from installed capacity / registry note
monthlyKgEstimate	Number	🕒 Estimated — tier midpoint formula
disposalMethod	String	✓ Real — from registry authorization type
source, sourceUrl	String	✓ Real — government PDF or company website URL
stage	Enum (5 stages)	★ Dummy — per PRD distribution rules
interest	Enum (High/Med/Low)	★ Dummy — probability by tier
lastContactDate	ISO date string	★ Dummy — days-ago formula by stage
attemptCount	Number (1–3)	★ Dummy — 3 if Qualified/Interested, else 1–2

4.3 Object 2 — Buyer lead (20 records)

Field	Type	Source
company, address, city, state	String	✓ Real — IndiaMART / company site / industry report
category	Enum (4 types)	✓ Real — SALT_MFR / CELL_MAKER / CAM_MAKER / METAL_RECOV
products	String[]	✓ Real — from IndiaMART listing or company product page
priceRef	Object {}	✓ Real — listed price from IndiaMART where published
consumptionMT	Number	★ Dummy — per category range (see rules)
currentSupplier	String	★ Dummy — 'China imports' / 'Chemical traders'
stage	Enum (5 stages)	★ Dummy — strategic buyers forced to Discussion
loiProbability	Number 0–100	★ Dummy — derived from stage
sampleRequested	Boolean	★ Dummy — true if stage ≥ Sample Requested

4.4 Object 3 — Collection point (32 records)

Field	Type	Source
name, address	String	✓ Real (17 entries); ★ Placeholder text (15)
contactPerson, phone	String null	✓ Real (Saahas published contacts); null otherwise
lat, lng	Number	✓ Google Maps from address (17); ① city-centre ± seeded offset (15)
city, state, category	String	✓ Real
nearestHub	String (hub ID)	① Derived from coordinates
monthlyKgEstimate	Number	① From category lookup table
willingness	Enum	★ Dummy — Yes for verified; Maybe for field
distanceToHubKm	Number	★ Seeded 10–70 km

4.5 Object 4 — Market price record (30 daily records)

Field	Source / Rule
date (Jun 7 – Jul 6 2026)	★ Generated sequence
cobalt_usd	✓ Real anchor \$56,290 + ★ seeded ±2%/day random walk (seed 31415)

Field	Source / Rule
nickel_usd	✓ Real anchor \$15,500 + ★ seeded $\pm 1.5\%$ /day (seed 27182)
liCarb_usd	✓ Real anchor \$9,271 + ★ seeded $\pm 1\%$ /day (seed 14142)
cobalt_inr / nickel_inr / liCarb_inr	● Derived: $\text{USD} \times \text{USD_INR} / 1000$ (gives INR per kg)
USD_INR	✓ Real ₹95.0 — exchange-rates.org Jun 2026

4.6 Dummy data generation rules (complete reference)

Field	Rule
Supplier pipeline stage	5 = Contacted; 3 forced = Qualified (TES-AMM, E-Parisaraa HQ, E-Parisaraa Chennai); 1 forced = Interested (Attero Bengaluru). Remainder: Lead or Contacted via seeded split.
Buyer pipeline stage	Epsilon, Amara Raja, Exide Energy = Discussion (forced). Pondy Oxides, Hydromet = Contacted. Strategic gigafactory makers (Ola, GODI, Tata) = Lead.
Monthly kg estimate	100–500 tier → 250 kg. 500–1000 → 750 kg. 1 Ton+ → 1,500 kg. Pan-India aggregators → 3,000 kg.
Last contact date	Qualified/Interested: seeded random(1–4) days ago. Contacted: random(2–8) days. Lead: random(9–30) days.
Interest level	1 Ton+ / 3 Ton+: 60% High, 25% Medium, 15% Low. 500–1000: 20% High, 50% Medium, 30% Low. 100–500: 15% High, 35% Medium, 50% Low.
Price time series	Each day = $\text{prev_day} \times (1 + (\text{rand}() - 0.5) \times 2 \times \text{vol_pct})$. Seeded per-metal. Same output every page load.
Buyer consumption MT/mo	Cell/gigafactory makers: 100–200 MT. CAM makers: 50–300 MT. Metal recovery: 30–40 MT. Salt mfrs: 3–50 MT.
Collection point lat/lng (field)	City centre $\pm (\text{rand}() - 0.5) \times 2 \times 0.04^\circ$. Seed 77. Deterministic offsets.

05

UX Design & Interactions

Navigation architecture, interaction patterns, loading states

5.1 Navigation architecture

Top bar	Full-width fixed header — logo left, 5 module tabs centre, pipeline health score ring + alert bell right
Active tab indicator	2px bottom border in module signature colour (blue M1, teal M2, amber M3, purple M4, red M5)
URL state	Filter selections update URL params (?city=Chennai&stage=Qualified) — bookmarkable and shareable
Mobile	Tabs collapse to dropdown on <768px. Charts reflow to single column. Map goes full-width.
No routing library	Single App.jsx with useState(activeTab) drives all navigation — no React Router needed

5.2 Full interaction map

Element	Trigger	Outcome
Funnel stage bar	Click	Filters lead table below to that pipeline stage only
City bar in chart	Click	Filters table and map to that city
Map pin (Leaflet)	Click	Opens info card: name, category, monthly volume, willingness, source
Metal price line point	Hover	Tooltip: date + USD price + INR equivalent + source citation
KPI card ⓘ icon	Hover	Data provenance tooltip: source name, URL, date recorded
Lead table row	Click	Row expands to show all fields for that lead (phone, email, source URL)
Margin calculator slider	Drag	Live-recalculates gross, cost, and net INR per kg
Chemistry dropdown (M4)	Change	Re-runs margin calculation with new recovery profile
AI chat input (M5)	Enter key	Fires Claude API call; response streams character-by-character
Export CSV button	Click	Downloads currently visible/filtered data as .csv via PapaParse
Alert bell (top bar)	Click	Slides in alerts panel from right (CSS transform, no fixed position)
Funnel comparison toggle (M2)	Click	Switches between single-funnel and side-by-side comparison view
Hub circle (M3 map)	Click	Highlights all collection points within that hub and shows economics sidebar

Element	Trigger	Outcome
Refresh prices button (M4)	Click	Updates the USD/INR assumption and recalculates all INR values

5.3 Loading and animation states

Event	Animation
Page initial load	Thin progress bar sweeps top of screen (1.0 s). KPI cards count up from 0 over 1.2 s via requestAnimationFrame.
Tab switch	Funnel bars animate width from 0 → final over 400 ms CSS transition. Chart lines draw in from left.
Map load (M3)	Pins drop with staggered 80 ms delay per pin using Leaflet marker add timing.
AI insight call (M5)	Spinner with 'Analysing your pipeline...' text. Response streams character-by-character.
Alert panel open	CSS transform translateX(100%) → 0 over 250 ms. Panel slides in from right.
Margin calculator update	Output values fade to grey, recompute, fade back to full opacity over 150 ms.

5.4 Responsive layout grid

Top bar: full width. KPI row: 4-column grid on desktop ($\geq 1024\text{px}$), 2×2 on tablet, 1-column on mobile. Content row: 60/40 split (chart left, secondary right) on desktop, stacked on tablet. Lead table: full width with horizontal scroll on mobile. All breakpoints handled by Tailwind responsive prefixes (md:, lg:).

06

Tech Stack & Deployment*Library choices, file structure, build commands***6.1 Recommended technology choices**

Layer	Technology	Reason
UI framework	React 18 with Vite	Vite is faster than CRA. HMR for rapid iteration. Evaluators expect React.
Charts	Recharts 2.x	Declarative React-native API. Supports all chart types needed (Line, Bar, Funnel, Pie). No canvas API.
Geo-map	Leaflet.js + react-leaflet	Free, no API key required, supports marker clusters, circle overlays, popup cards.
Styling	Tailwind CSS (CDN v3)	Utility-first. Professional appearance with zero custom CSS writing. Responsive built-in.
Data layer	Static JS files in /src/data/	All 4 data files pre-generated. No backend, no API call needed for pipeline data.
AI integration	Anthropic API (claude-sonnet-4-20250514)	Called from frontend fetch(). No proxy needed for assessment context.
CSV export	PapaParse	Lightweight. Browser download via URL.createObjectURL(). No server needed.
Deployment	Vercel (free tier)	Auto-deploy on git push. Public URL live in under 5 minutes. Free forever for this use case.

6.2 Project file structure

The structure below reflects the component architecture derived from the 5-module PRD. Every component maps to exactly one module or one reusable piece.

```

jol-dashboard/
  src/
    components/
      SupplierPipeline.jsx    ← Module 1
      BuyerPipeline.jsx      ← Module 2
      CollectionNetwork.jsx   ← Module 3
      MarketIntelligence.jsx  ← Module 4
      AIInsights.jsx          ← Module 5
      KPICard.jsx             ← Reusable metric card with ① tooltip
      FunnelChart.jsx         ← Reusable funnel (supplier + buyer)
      HealthScore.jsx         ← Ring gauge composite score
      AlertsPanel.jsx         ← Smart alerts slide-in drawer
      MarginCalculator.jsx     ← Module 4 interactive widget
      LeadTable.jsx           ← Sortable/filterable data table
      GeoMap.jsx              ← Leaflet map with hub overlay
    data/
      supplierLeads.js        ← 40 real + dummy-filled leads ✓ GENERATED
      buyerLeads.js          ← 20 real + dummy-filled buyers ✓ GENERATED

```



```

    collectionPoints.js    ← 32 points + lat/lng + hubs    ✓ GENERATED
    marketPrices.js       ← 30-day series + calc engine  ✓ GENERATED
  utils/
    healthScore.js        ← Composite score formula
    exportCsv.js           ← PapaParse download helper
    formatters.js         ← INR formatting, date helpers
  App.jsx                 ← Tab navigation + global state
  main.jsx                ← Vite entry point
public/
  package.json
  vite.config.js
  tailwind.config.js
  vercel.json             ← { rewrites: [{ source:'/(.*)', dest:'/index.html'
}} ]

```

6.3 Installation and deploy commands

```

# 1. Scaffold project
npm create vite@latest jol-dashboard -- --template react
cd jol-dashboard && npm install

# 2. Install dependencies
npm install recharts leaflet react-leaflet papaparse
npm install -D tailwindcss postcss autoprefixer
npx tailwindcss init -p

# 3. Copy generated data files
cp /path/to/supplierLeads.js src/data/
cp /path/to/buyerLeads.js src/data/
cp /path/to/collectionPoints.js src/data/
cp /path/to/marketPrices.js src/data/

# 4. Add Anthropic API key to .env
echo 'VITE_ANTHROPIC_KEY=sk-ant-...' >> .env

# 5. Run locally
npm run dev → http://localhost:5173

# 6. Deploy to Vercel (takes ~5 minutes)
git init && git add . && git commit -m 'Jol Energy dashboard v1'
# → Connect repo at vercel.com → Import → Deploy
# → Public URL: https://jol-dashboard-[yourname].vercel.app

```

6.4 Key implementation notes

- The Anthropic API key must be in .env as VITE_ANTHROPIC_KEY. Vite exposes it to the browser via import.meta.env. For a production app you would proxy through a server — for this assessment, browser-direct is acceptable.
- The Leaflet CSS must be imported in main.jsx: import 'leaflet/dist/leaflet.css'. Without this the map tiles render but controls are invisible.

- react-leaflet v4 requires React 18. Use MapContainer, TileLayer, Marker, Popup, Circle from 'react-leaflet'. For clustering, add leaflet.markercluster separately.
- Recharts FunnelChart renders the funnel shape natively. Alternatively, build a custom SVG funnel with proportional-width rect elements for more visual control.
- For the animated count-up KPIs, use a custom hook useCountUp(target, duration) that steps from 0 to target via requestAnimationFrame over the specified duration in milliseconds.
- PapaParse CSV export: Papa.unparse(filteredData) returns a CSV string. Trigger download with URL.createObjectURL(new Blob([csv], {type:'text/csv'})) and a programmatic <a> click.

07

Submission Checklist*Final quality gate before the Google Form deadline*

The task brief says submissions without proof of work receive significantly lower scores. The checklist below ensures every evaluation criterion is covered before you submit.

7.1 Required Google Form uploads

#	File	Format	What to verify before uploading
1	Presentation	PPT or PDF, 10–15 slides	All 10 required slides present; slide order matches task PDF exactly
2	Lead database	Excel (.xlsx)	40 supplier leads, 20 buyer leads, 32 collection points, 30-day prices; all columns filled
3	Evidence folder	Zipped .zip	Email screenshots, call log screenshots, geo-tagged map screenshots, any WhatsApp
4	Dashboard	Vercel URL or .pbix file	Vercel URL loads without console errors; all 5 modules render; data matches Excel
5	Insights report	PDF or Word (.docx)	2–3 pages; every number cited; 3 specific recommendations; no spelling errors

7.2 Pre-submission quality checklist

1. Every number in the dashboard has a source — verify ⓘ tooltips appear on all KPI cards.
2. Lead database has no blank required fields — check interest level and pipeline stage for all 40 rows.
3. At least one screenshot of a sent email or WhatsApp message is in the evidence folder.
4. Dashboard loads on Chrome and Edge without console errors.
5. Presentation follows exact slide order from the task PDF.
6. Metal prices are dated — not just 'current price' with no date.
7. Insights report contains 3 specific, numbered recommendations — not generic bullet points.
8. All files named clearly: JolEnergy_Leads_[Name].xlsx, JolEnergy_Dashboard_[Name].pdf, etc.
9. The Vercel URL is public — verify by opening in an incognito window before submitting.
10. README.md in the repository explains how to update the data files and refresh the dashboard.

7.3 Scoring maximisation summary

Criterion	Weight	Maximum score action
Real effort & execution	35%	Deployed Vercel URL + real TNPCB/KSPCB phone numbers + Claude API integration + data provenance on every metric
Business opportunities	25%	40 verified supplier leads + 20 buyer leads + margin calculator + AI city recommendation engine
Proof of work	20%	Source URL on every data row + email/WhatsApp screenshots in evidence folder + geo-tagged map screenshots
Data quality & accuracy	10%	Every metric traces back to government registry, IndiaMART listing, or SMM/Trading Economics with citation date
Presentation & communication	10%	Insights report with 3 specific data-backed claims + clean dashboard design + polished 12-slide PPT

The single highest-impact action in this entire assessment: deploy to Vercel and submit the URL. No other SDE intern submission will include a live, publicly accessible dashboard application. That URL alone puts the submission in the top tier.